

Winning Space Race with Data Science

Janani G



Outline



01

Executive Summary

02

Introduction

03

Methodology

04

Results

05

Conclusion

06

Appendix

- The following methodologies were used to analyze data:
- Data Collection using web scraping and SpaceX API;
- Exploratory Data Analysis (EDA), including data wrangling, data visualization and interactive visual analytics;
- Machine Learning Prediction.

Summary of all results

- It was possible to collect valuable data from public sources;
- EDA allowed to identify which features are the best to predict success of launchings;
- Machine Learning Prediction showed the best model to predict which characteristics are important to drive this opportunity by the best way, using all collected data.

Introduction



SpaceX's goal:

- Sending spacecraft to the International Space Station.
- Starlink, a satellite internet constellation providing satellite Internet access.
- Sending manned missions to Space.

Object

SpaceX's Falcon 9 launch

Goal

To determine the price of each launch.

Reason

If we can determine if the first stage will land, we can determine the cost of a launch

Method

By gathering information about Space X and creating dashboards:

- To determine if SpaceX will reuse the first stage
- Train a machine learning model and use public information to predict if SpaceX will reuse the first stage

Section 1

Methodology

- Data collection methodology:
 - Require the data from SpaceX API
 - Collect data from a Wikipedia page
- Perform data wrangling
 - Perform EDA to find some patterns
 - Determine what would be the label for training supervised model

Perform exploratory data analysis (EDA) using visualization and SQL

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Compare logistic regression model, support vector machine tree decision classifier, KNN by using GridSearchCV to select the best fit model

Data Collection



1 AP

spacex_url="<https://api.spacexdata.com/v4/launches/past>"

- Required the data from Space API
- Clean the data

2 Web

https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

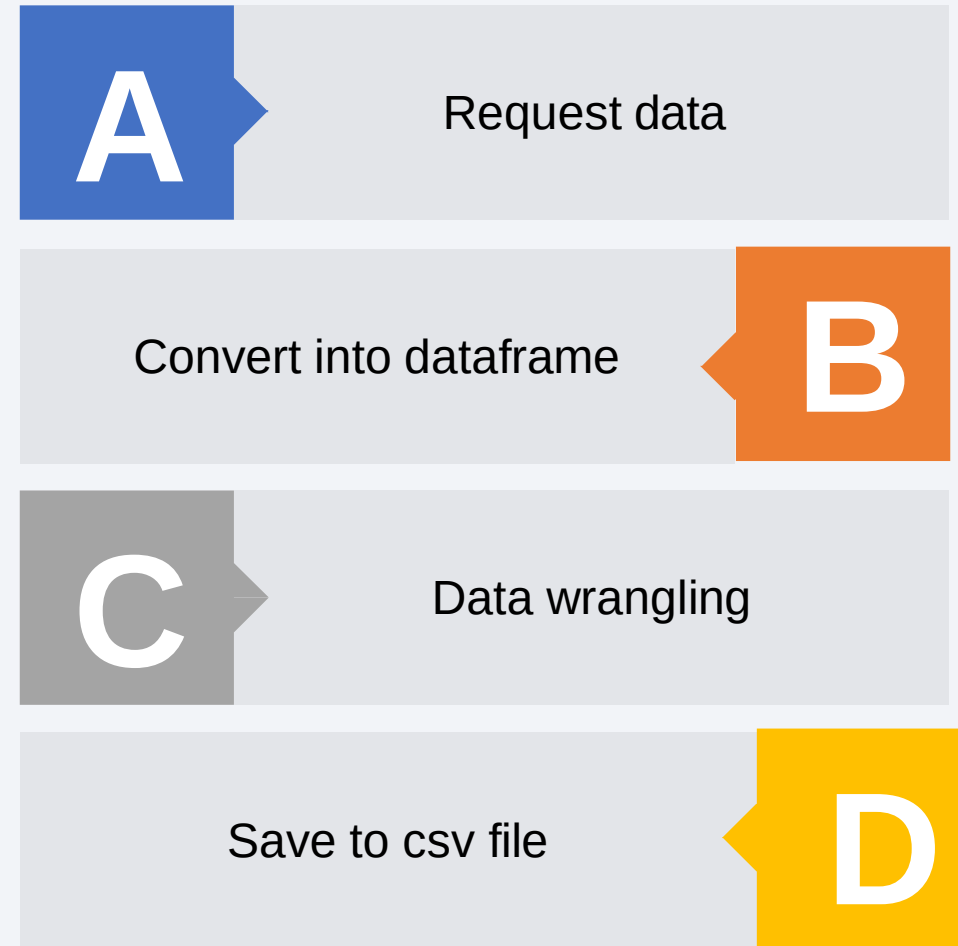
- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

Data Collection - SpaceX API



- Request data from SpaceX API
- Convert the json result into a dataframe
- Filter dataframe to only Falcon 9 launches and data wrangling
- Export to csv
- Link :

<https://github.com/Jananijana2712/SpaceX-Falcon-9-First-Stage-Landing-Prediction/blob/main/1.1%20SpaceX%20the%20Data%20Collection%20API%20Lab.ipynb>

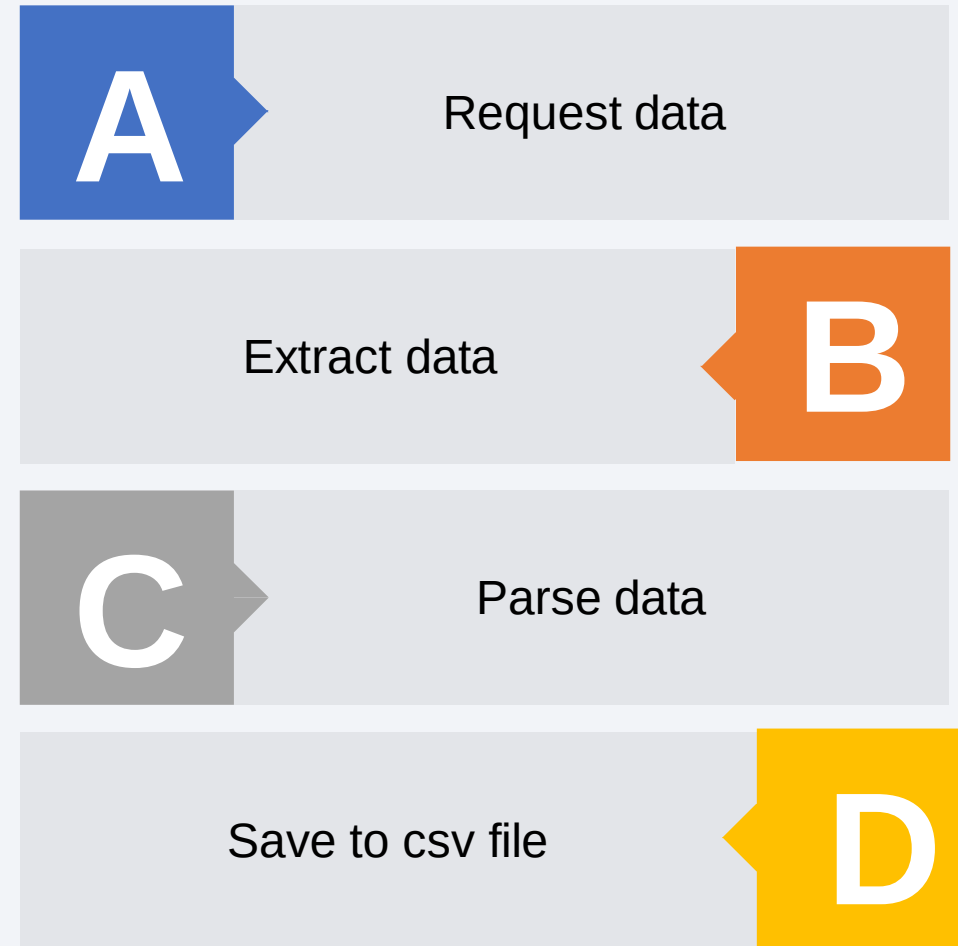


Data Collection - Scraping



- Request the Falcon9 Launch Wiki page from its URL
- Extract all column/variable names from the HTML table header
- Create a data frame by parsing the launch HTML tables
- Export to csv
- Link :

https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.2%20SpaceX%20the%20Data%20Collection%20with%20Web%20Scraping%20lab.ipynb

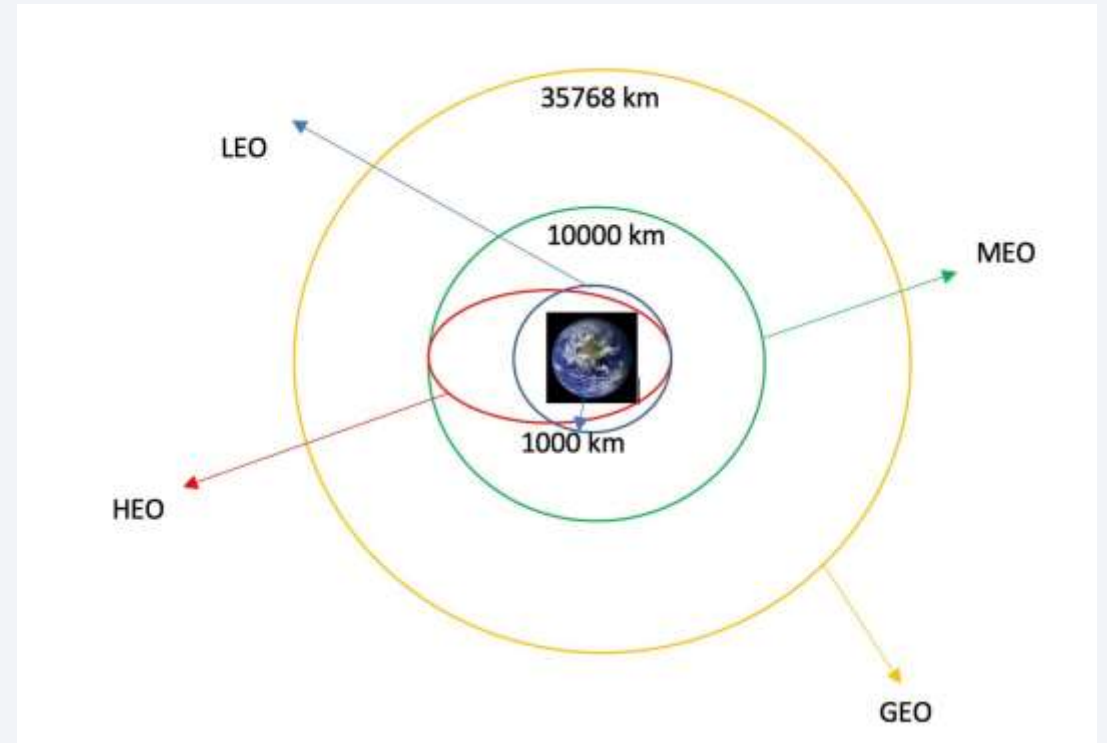


Data Wrangling



- Specify the missing value
- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of each orbit
- Create a landing outcome label from Outcome column

Link: https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.3%20SpaceX%20Data%20Wrangling.ipynb



- The relationship between Flight Number and Launch Site -> scatter plot
 - The relationship between Payload and Launch Site -> scatter plot
 - The relationship between success rate of each orbit type -> bar plot
 - The relationship between Flight Number and Orbit type -> scatter plot
 - The relationship between Payload and Orbit type -> scatter plot
 - The launch success yearly trend -> line chart
-
- The scatter plot is the best to describe the relation between two categorical data
 - The bar plot is the best to compare several categorical data
 - The line plot is the best to show the time series data



https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.4%20SpaceX%20%20the%20EDA%20with%20Visualization.ipynb

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster_versions which have carried the maximum payload mass.
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Rank the count of successful landing_outcomes between the date 04-06-2010 and 20-03-2017 in descending order.



https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/Assignment%20SQL%20Notebook%20for%20Peer%20Assignment.ipynb

Build an Interactive Map with Folium



- Mark all launch sites on a map
- Mark the success/failed launches for each site on the map
- Calculate the distances between a launch site to its proximities
 - Whether it is close to the coast
 - Whether it is close to the railway
 - Whether it is close to the highway
 - Whether it is close to the city

▶ **To find some geographical patterns about launch sites**



https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.5%20SpaceX%20Interactive%20Visual%20Analytics%20with%20Folium.ipynb

Build a Dashboard with Plotly Dash



- A launch site drop-down input component
- A success-pie-chart based on the selected site dropdown
- A range slicer to select payload
- A success-payload-scatter-chart scatter plot based on the selected site dropdown

► **To inspect the relationship of success rate between launch site and payload**



https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction/blob/main/1.6%20SpaceX%20Machine%20Learning%20Prediction.ipynb

Predictive Analysis (Classification)



Data wrangling

Data standarization

Split into traning
and test datasets

Predictive
model
evalutation

Predictive
model
selection



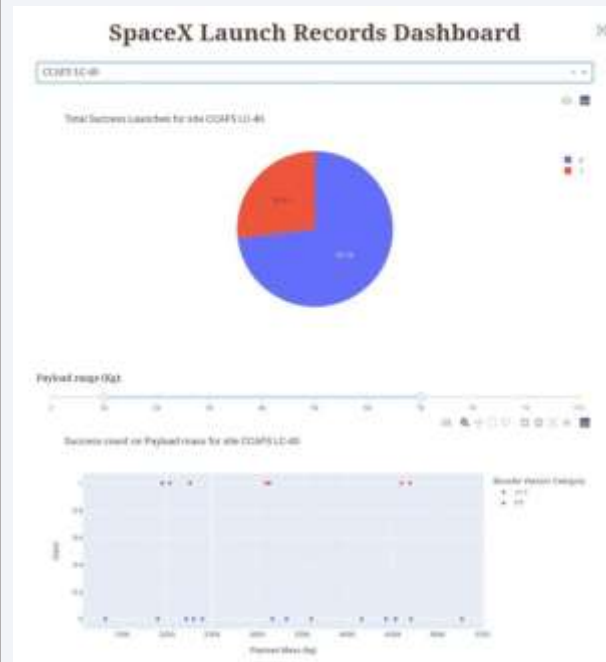
https://github.com/xiaojie-qian/SpaceX-Falcon-9-first-stage-Landing-Prediction/blob/main/1.7%20SpaceX_Machine%20learning%20predictive%20analysis.ipynb

- Logistic regression
- Support vector machine
- Decision tree classifier
- K-nearest neighbors

EDA

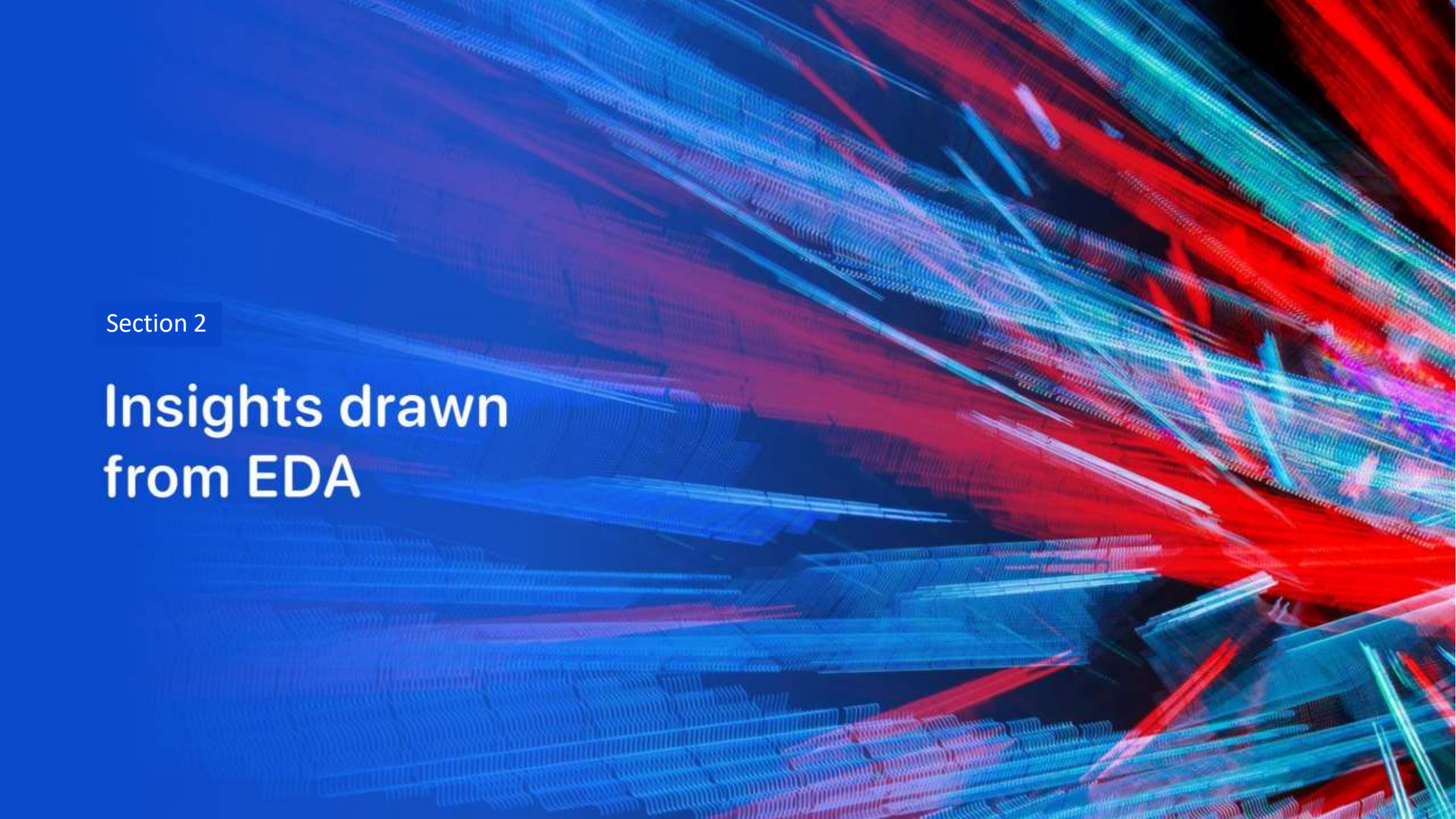
- KSC LC-39A and VAFB SLC 4E has a success rate of 77%
- VAFB SLC 4E has no payload above 10000 kg
- In the LEO orbit the Success appears related to the number of flights
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS
- The success rate since 2013 kept increasing till 2020

Interactive analytics



Predictive analysis

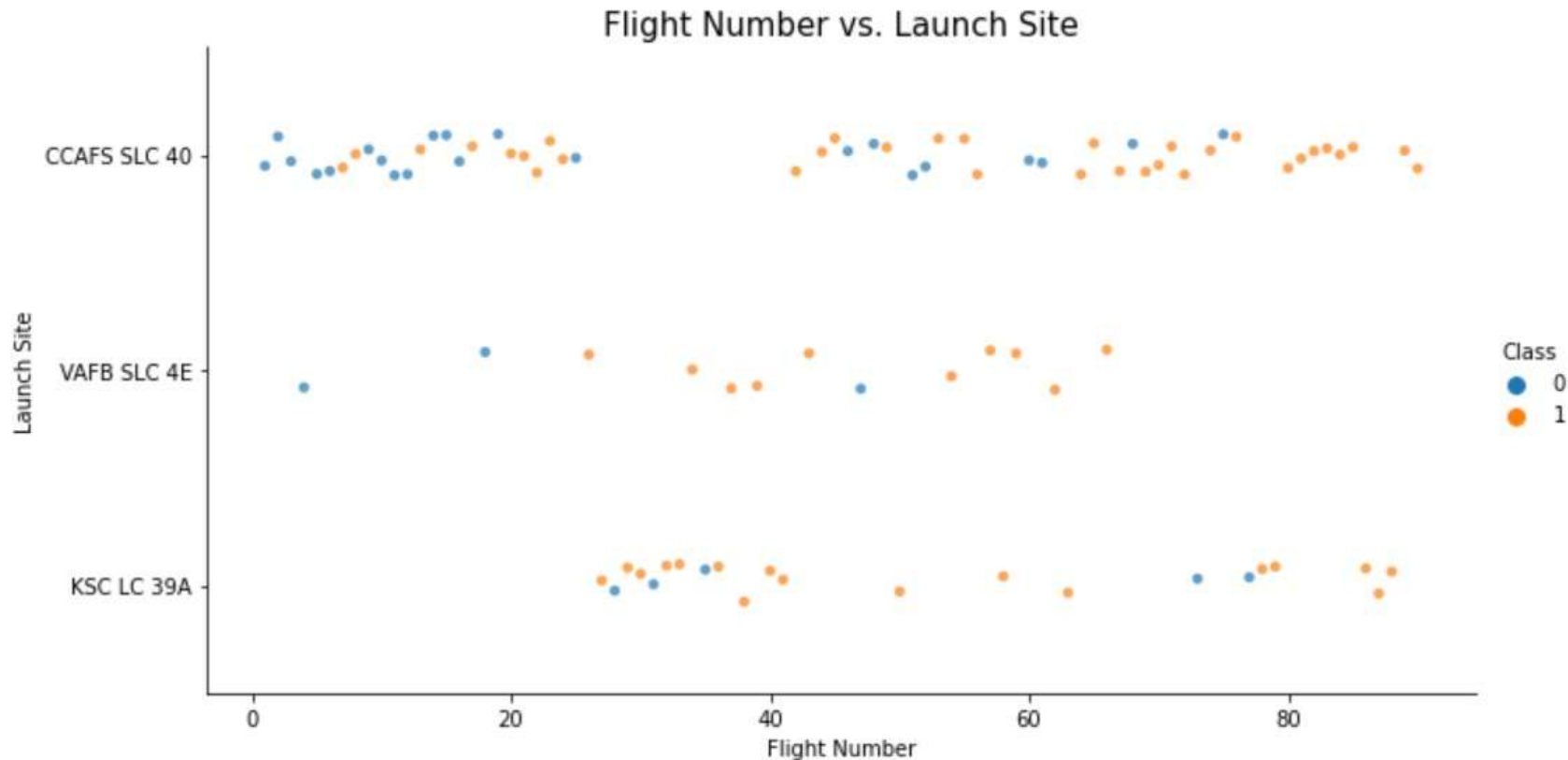
According to the decision tree classifier model, the predictive model tells us that there will be 4 true positive, 7 true negative, 5 false positive and 2 false negative. The accuracy of the model is around 89% with the best parameters.

The background of the slide is an abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and teal. These lines are oriented diagonally, creating a sense of motion and depth. The overall effect is a vibrant, digital-looking texture.

Section 2

Insights drawn from EDA

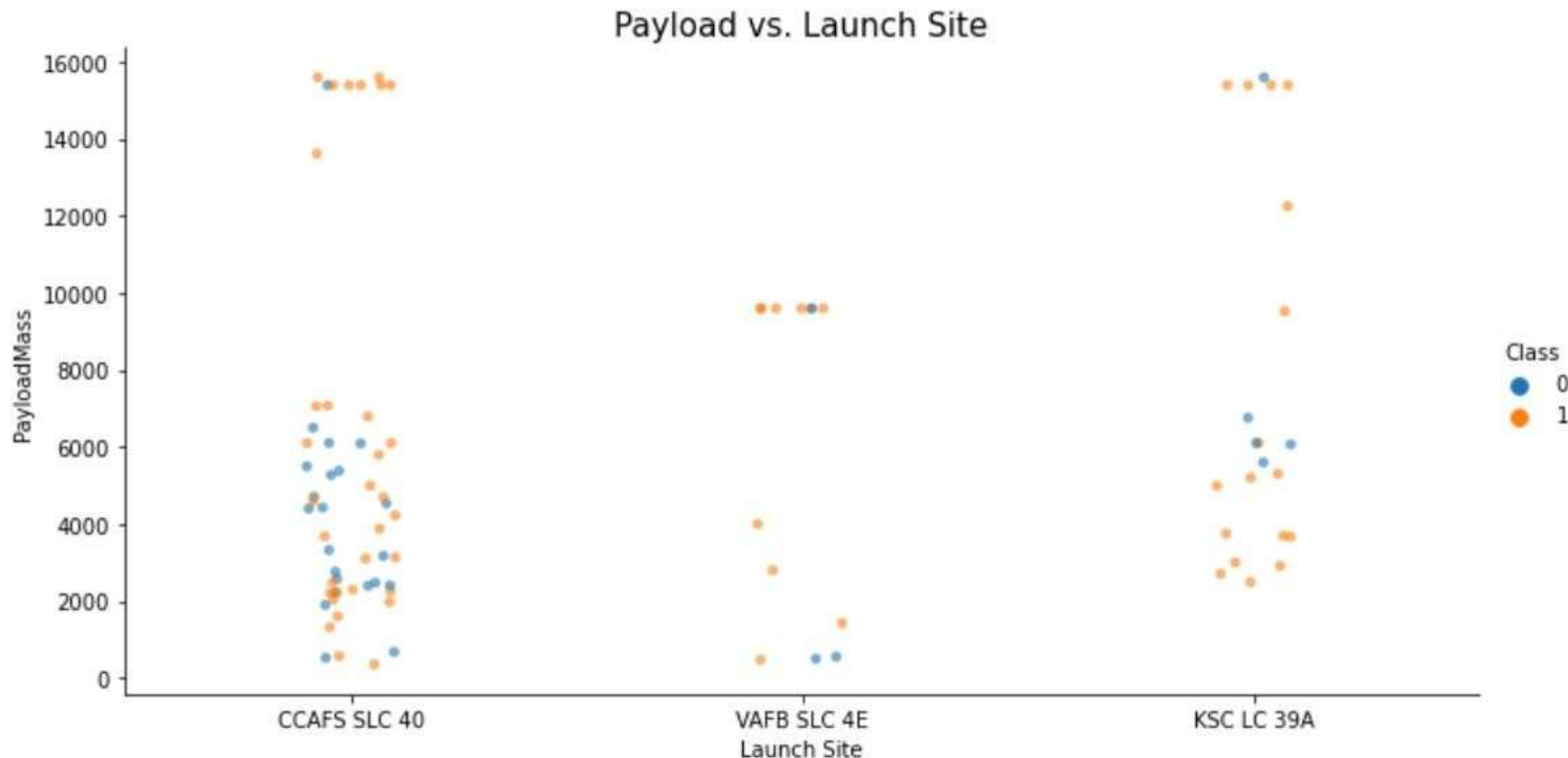
Flight Number vs. Launch Site



-> KSC LC 39A has the highest successful rate, vice versa CCAFS SLC 40 has the lowest

- According to the plot above, it's possible to verify that the best launch site nowadays is CCAFS SLC 40, where most of recent launches were successful.
- In second place VAFB SLC 4E and third place KSC LC 39A.
- It's also possible to see that the general success rate improved over time.

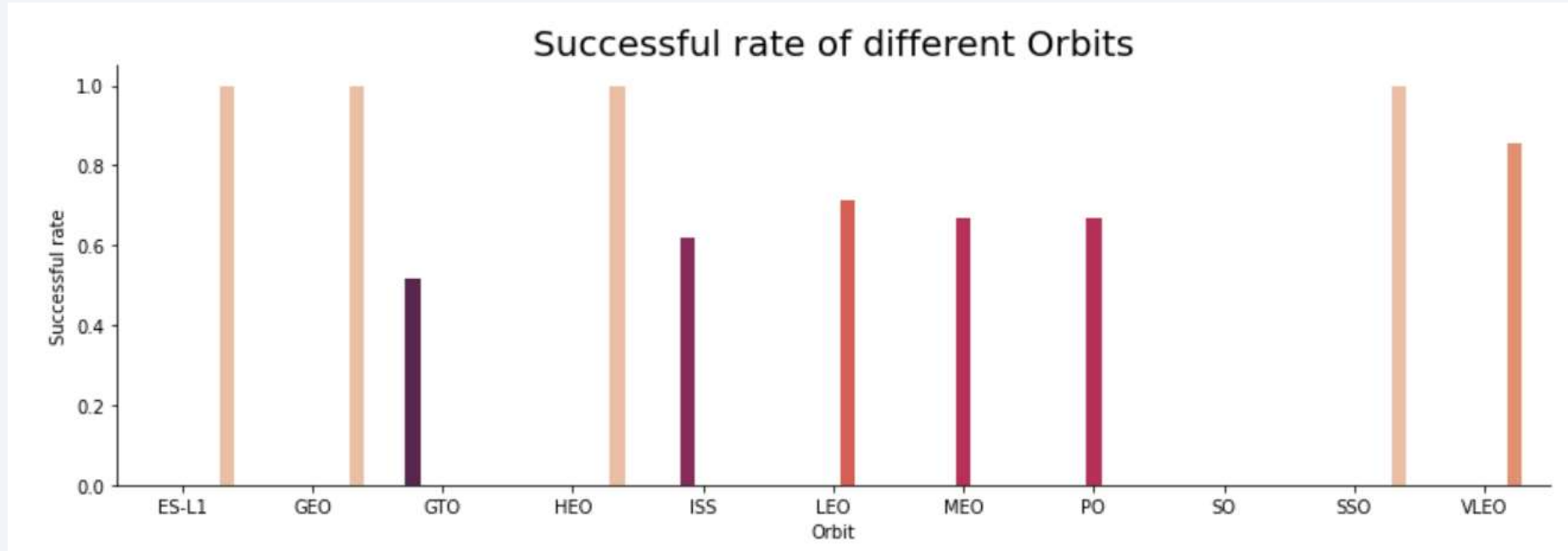
Payload vs. Launch Site



-> VAFB SLC 4E has no payload above 10000 kg

- Payloads over 9,000kg (about the weight Of a school bus) have excellent success Rate.
- Payloads over 12,000kg seems to be possible only on CCAFS SLC 40 and KSC LC 39A launch sites.

Success Rate vs. Orbit Type



-> ES – L1, GEO, HEO, SSO have the highest success rate

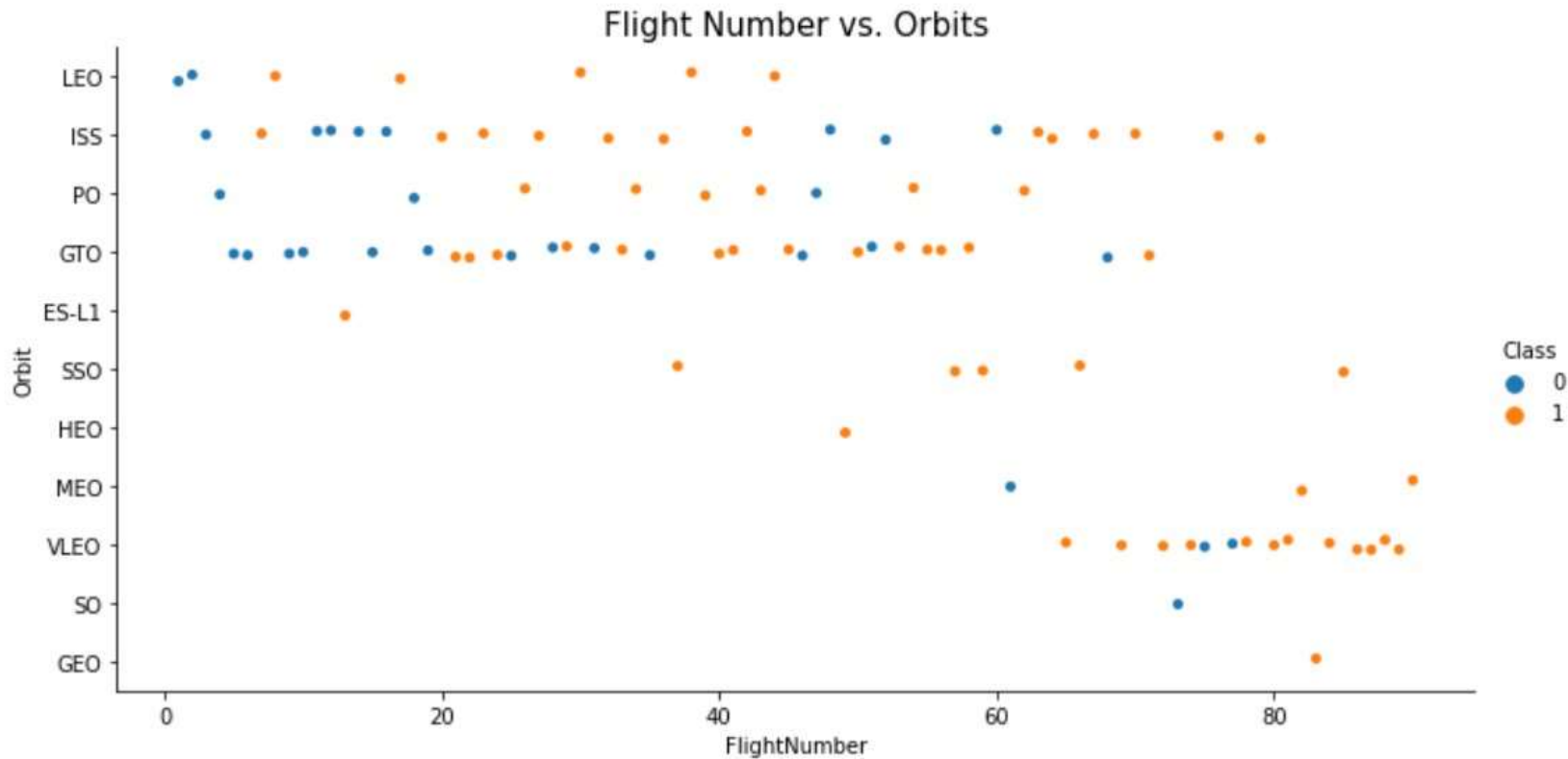
The biggest success rates happens to orbits:

- ES-L1;
- GEO;
- and
- SSO.

Followed by:

- VLEO (above 80%) and
- LFO (above 70%).

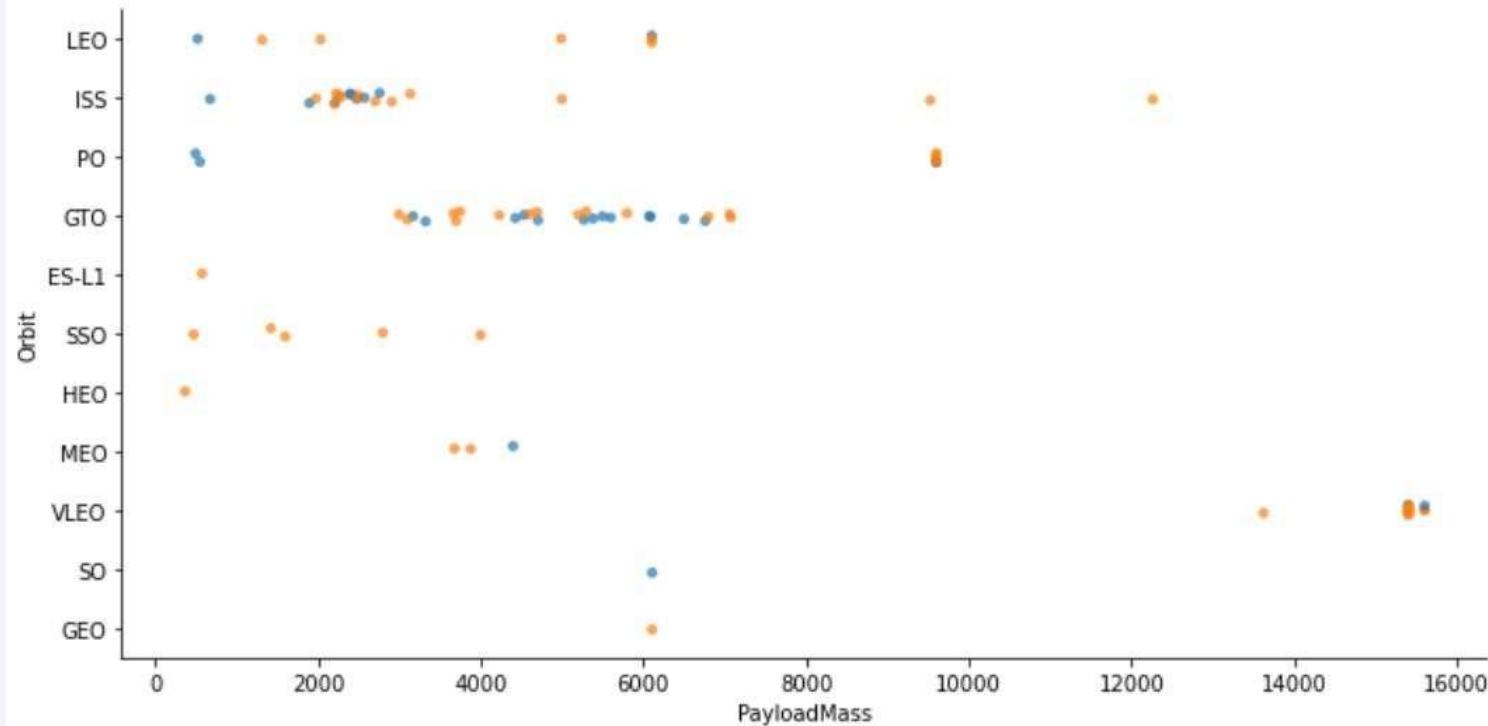
Flight Number vs. Orbit Type



- Apparently, success rate improved over time to all orbits;
- VLEO orbit seems a new business opportunity, due to recent increase of its frequency.

-> In the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

Payload vs. Orbit Type



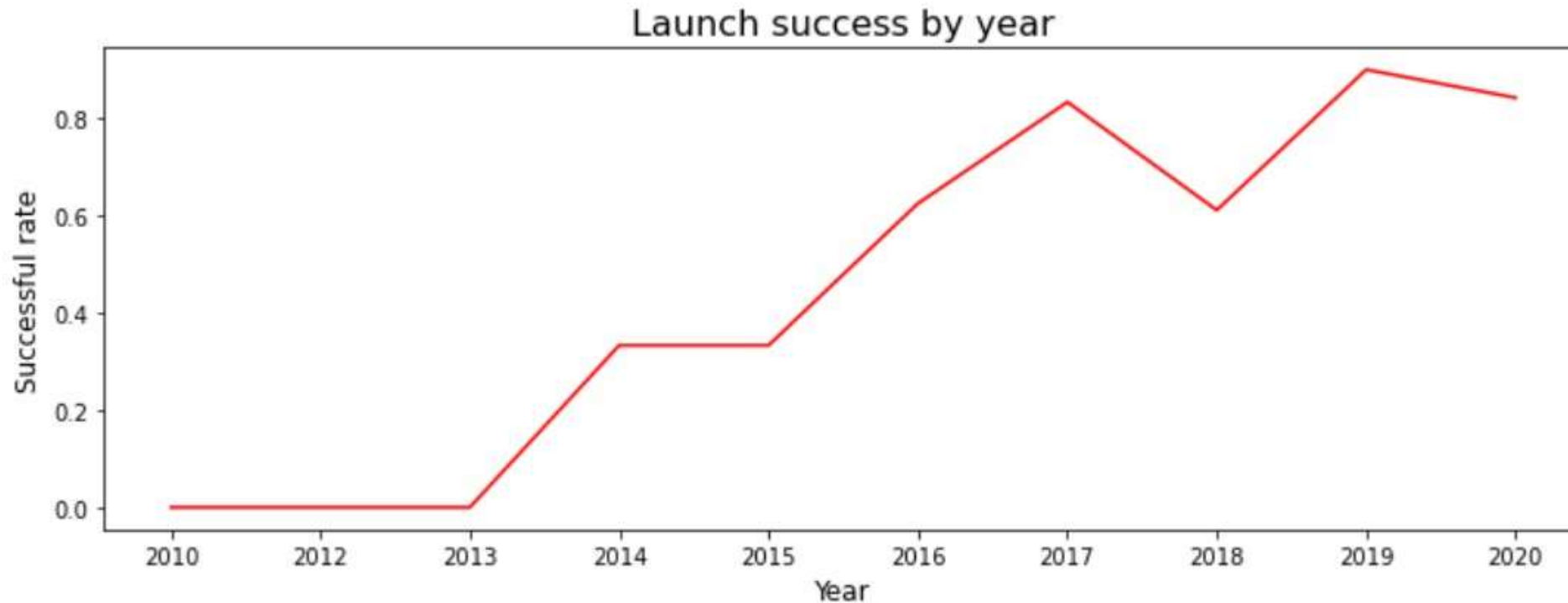
- Apparently, there is no relation between payload and success rate to orbit GTO.

- ISS orbit has the widest range Of payload and a good rate Of success.

- There are few launches to the orbits SO and GEO.

->With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccesful mission) are both there here.

Launch Success Yearly Trend



- Success rate started increasing in 2013 and kept until 2020.

- It seems that the first three years were a period of adjusts and improvement of technology.

-> The success rate since 2013 kept increasing till 2020

All Launch Site Names



According to data, there are four launch sites:

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

They are obtained by selecting unique occurrences of "launch_site" values from the dataset.

Launch Site Names Begin with 'CCA'



5 records where launch sites begin with 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Time JTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
5:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
3:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
4:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
5:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
0:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Here we can see five samples of Cape Canaveral launches.

Total Payload Mass



Total payload carried by boosters from NASA:

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload FROM SPACEXTBL WHERE Customer LIKE 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
total_payload
```

```
45596
```

Total payload calculated above, by summing all payloads whose codes contain 'CRS', which corresponds to NASA.

Average Payload Mass by F9 v1.1



Average payload mass carried by booster version F9 vl. 1 :

```
%sql SELECT avg(PAYLOAD_MASS_KG_) AS Avg_Payload FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Avg_Payload
```

```
2928.4
```

Filtering data by the booster version above and calculating the average payload mass we obtained the value of 2,928 kg.

First Successful Ground Landing Date



First successful landing outcome on ground pad:

```
%sql SELECT min(date) AS Early_Date from SPACEXTBL where Landing_Outcome LIKE 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Early_Date
```

```
01-05-2017
```

By filtering data by successful landing outcome on ground pad and getting the minimum value for date it's possible to identify the first occurrence, that happened on 12/22/2015.

Successful Drone Ship Landing with Payload between 4000 and 6000



Boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT Customer, Landing_Outcome, PAYLOAD_MASS__KG_ FROM SPACEXTBL  
WHERE Landing_Outcome ='Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db  
Done.
```

Customer	Landing_Outcome	PAYLOAD_MASS__KG_
SKY Perfect JSAT Group	Success (drone ship)	4696
SKY Perfect JSAT Group	Success (drone ship)	4600
SES	Success (drone ship)	5300
SES EchoStar	Success (drone ship)	5200

Selecting distinct booster versions according to the filters above, these 4 are the result.

Total Number of Successful and Failure Mission Outcomes



Number of successful and failure mission outcomes:

```
%sql SELECT Mission_Outcome, Count(*) AS Numbers FROM SPACEXTBL GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db  
Done.
```

Mission_Outcome	Numbers
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Grouping mission outcomes and counting records for each group led us to the summary above.

Boosters Carried Maximum Payload



Boosters which have carried the maximum payload mass

```
%sql SELECT Booster_Version, Max_Payload FROM (SELECT Booster_Version, MAX(PAYLOAD_MASS_KG_)
AS Max_Payload FROM SPACEXTBL GROUP BY Booster_Version)
```

* sqlite:///my_data1.db
Done.

Booster_Version	Max_Payload
F9 B4 B1039.2	2647
F9 B4 B1040.2	5384
F9 B4 B1041.2	9600
F9 B4 B1043.2	6460
F9 B4 B1039.1	3310
F9 B4 B1040.1	4990
F9 B4 B1041.1	9600
F9 B4 B1042.1	3500
F9 B4 B1043.1	5000
F9 B4 B1044	6092

These are the boosters which have carried the maximum payload mass registered in the dataset.

2015 Launch Records



Failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql SELECT SUBSTR(Date,4,2) AS Month, Booster_Version, Launch_site FROM SPACEXTBL  
WHERE Landing_Outcome LIKE 'Failure%drone%' AND SUBSTR(Date,7,4) = '2015'
```

```
* sqlite:///my_data1.db  
Done.
```

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

The list above has the only two occurrences.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20



Ranking of all landing outcomes between the date 2010-06-04 and 2017- 03-20:

```
%sql SELECT Landing_Outcome, COUNT(*) AS Numbers FROM SPACEXTBL  
WHERE Landing_Outcome LIKE 'Success%' AND Date BETWEEN '04-06-2010' AND '20-03-2017'  
GROUP BY Landing_Outcome ORDER BY Numbers DESC;
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	Numbers
Success	20
Success (drone ship)	8
Success (ground pad)	6

This view Of data alerts us that "NO attempt" must be taken in account.

A satellite view of Earth from space, showing the curvature of the planet and the glowing city lights of the Eastern United States and parts of Canada at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

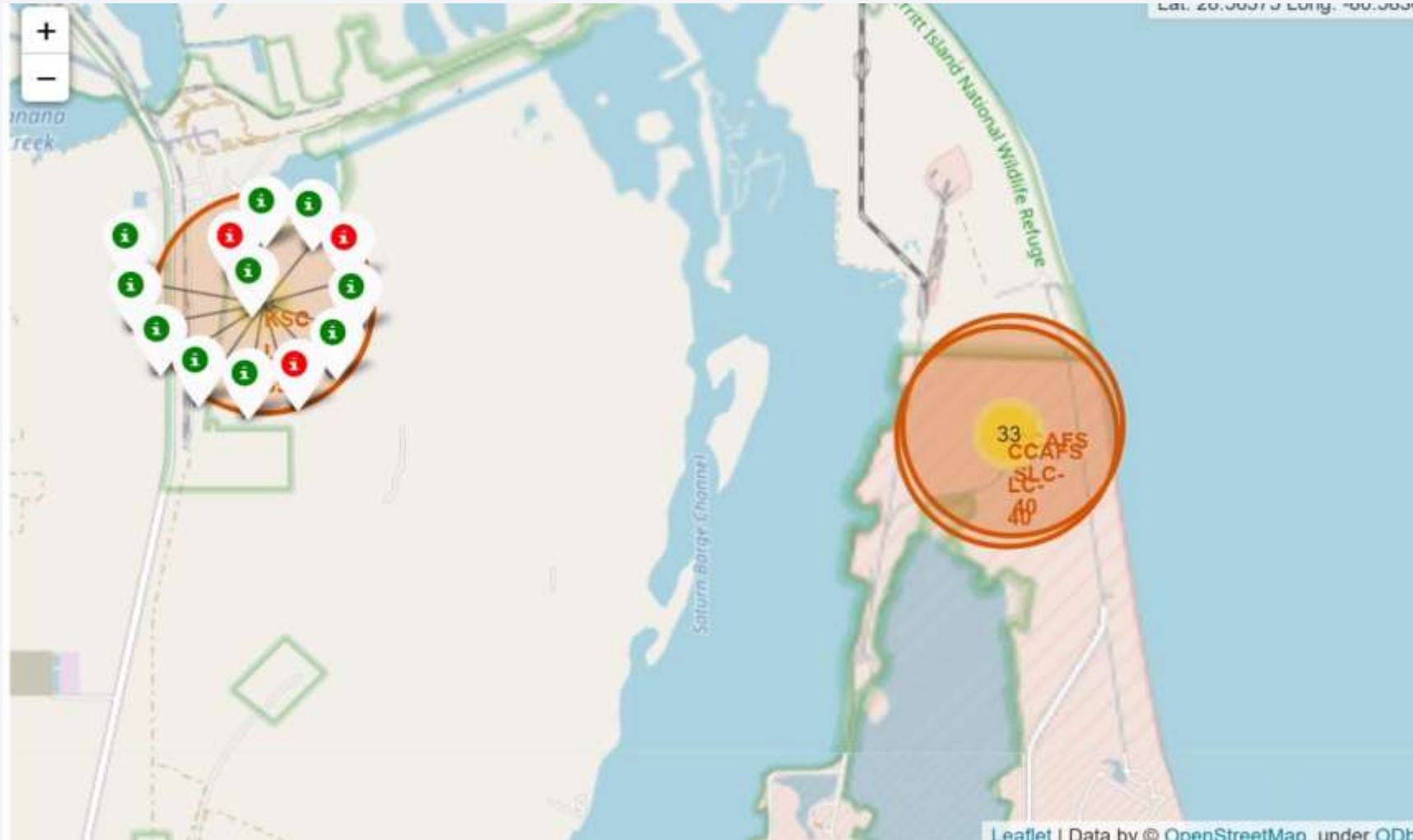
<All launch sites>



Launch sites are near sea, probably by safety, but not too far from roads and railroads.

- All launch sites in proximity to the Equator line
- All launch sites in very close proximity to the coast

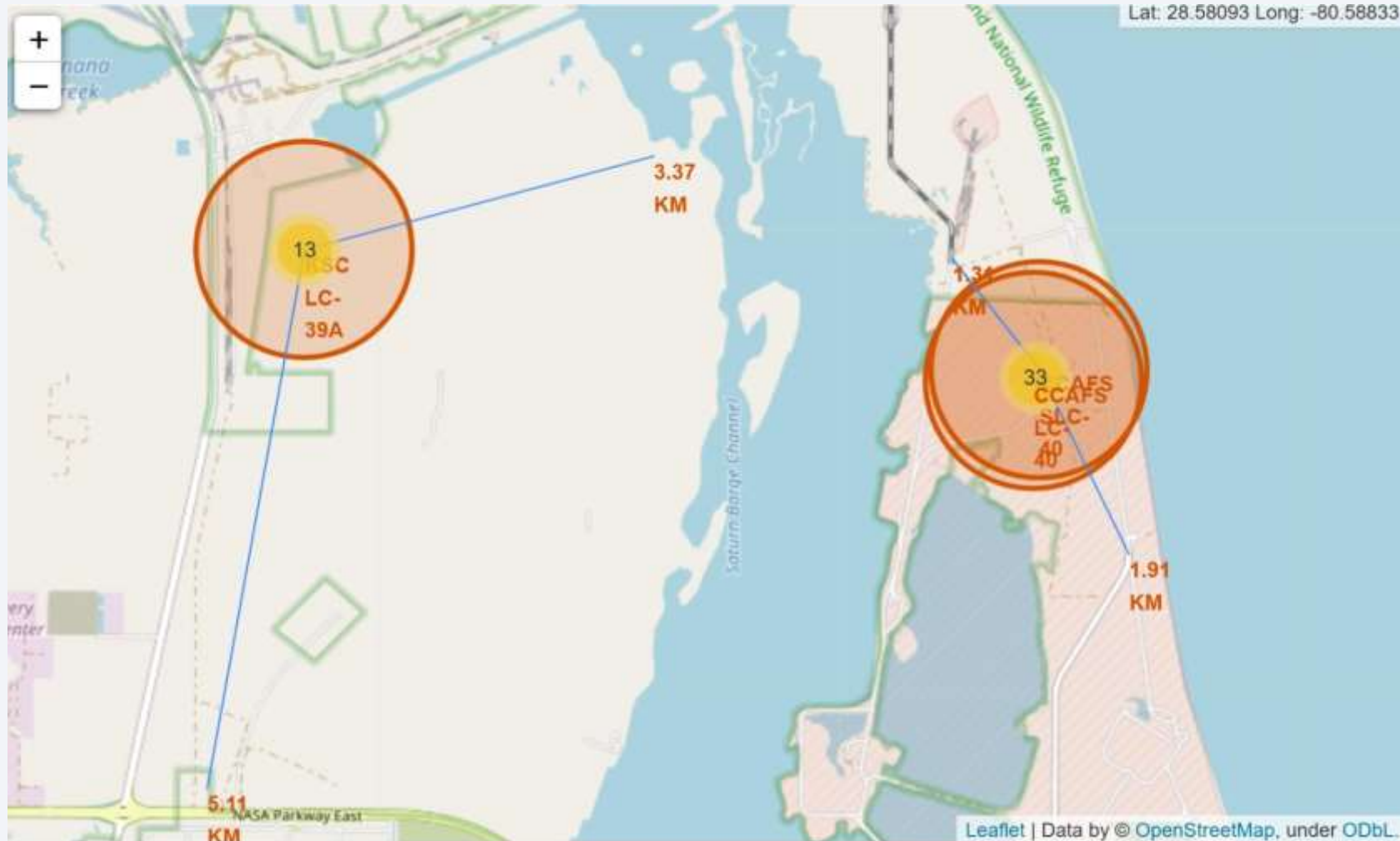
Launch outcome of different site



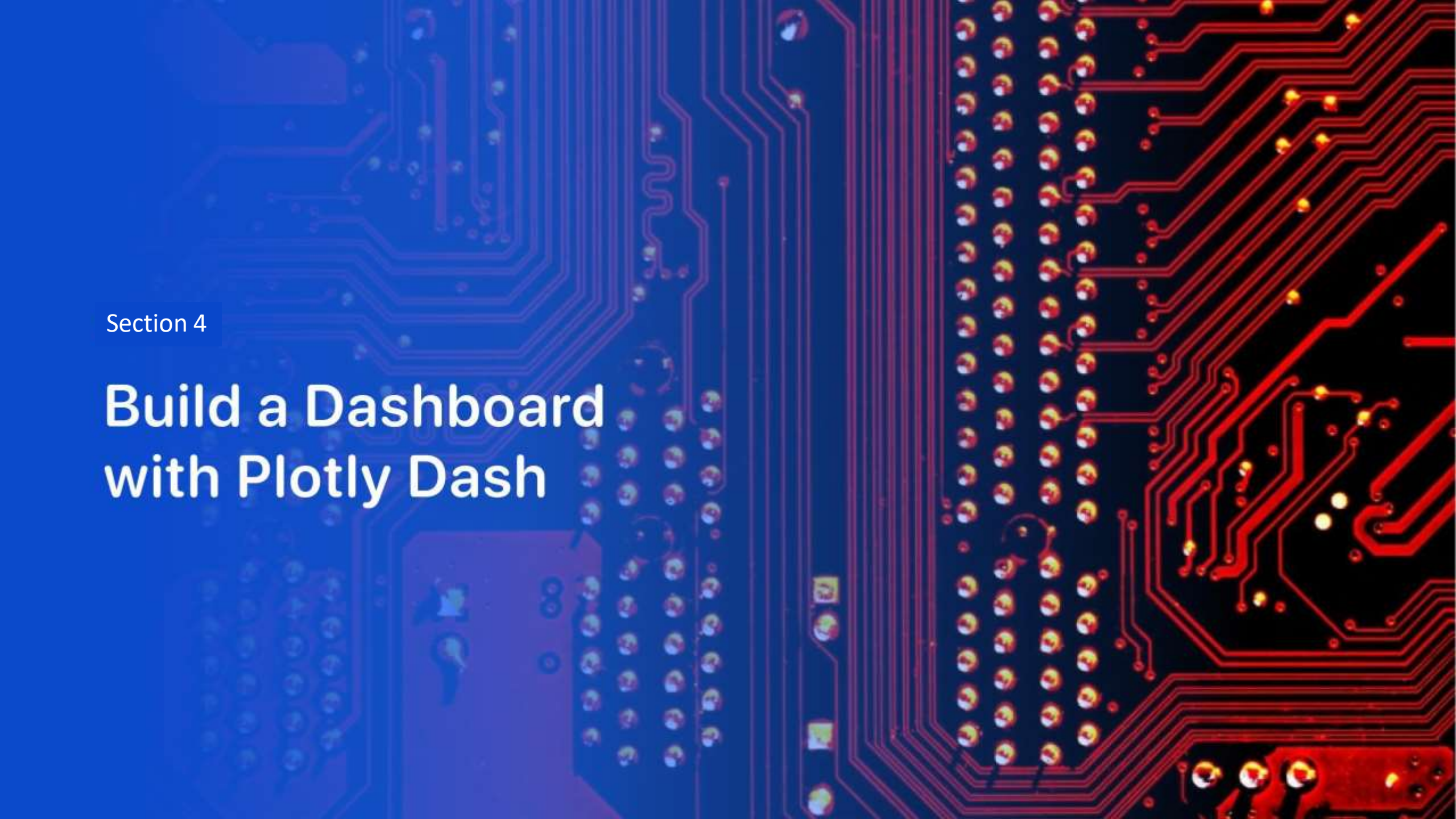
Example Of KSC
LC-39A launch site
launch outcomes

Green markers
indicate
successful and
red ones indicate
failure.

The proximity of the launch sites



Launch site KSC LC-39A has good logistics aspects, being near railroad and road and relatively far from inhabited areas.



Section 4

Build a Dashboard with Plotly Dash

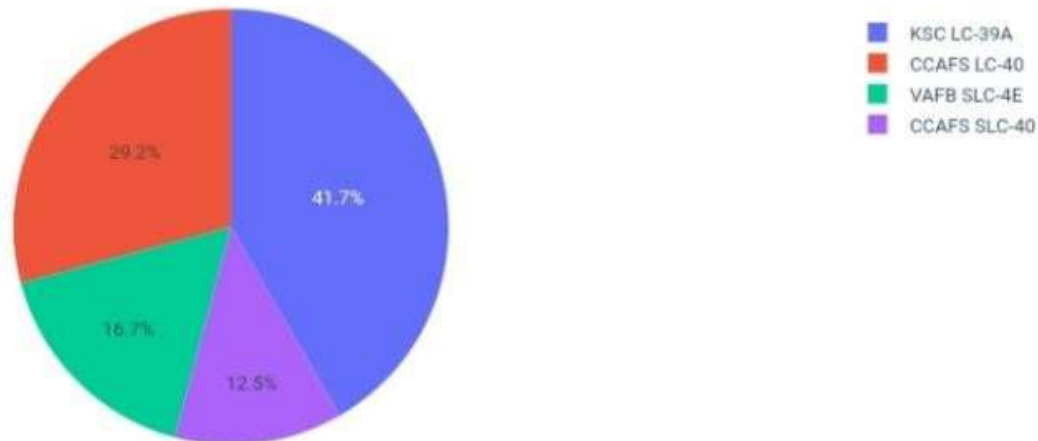
All site launch



SpaceX Launch Records Dashboard

All Sites

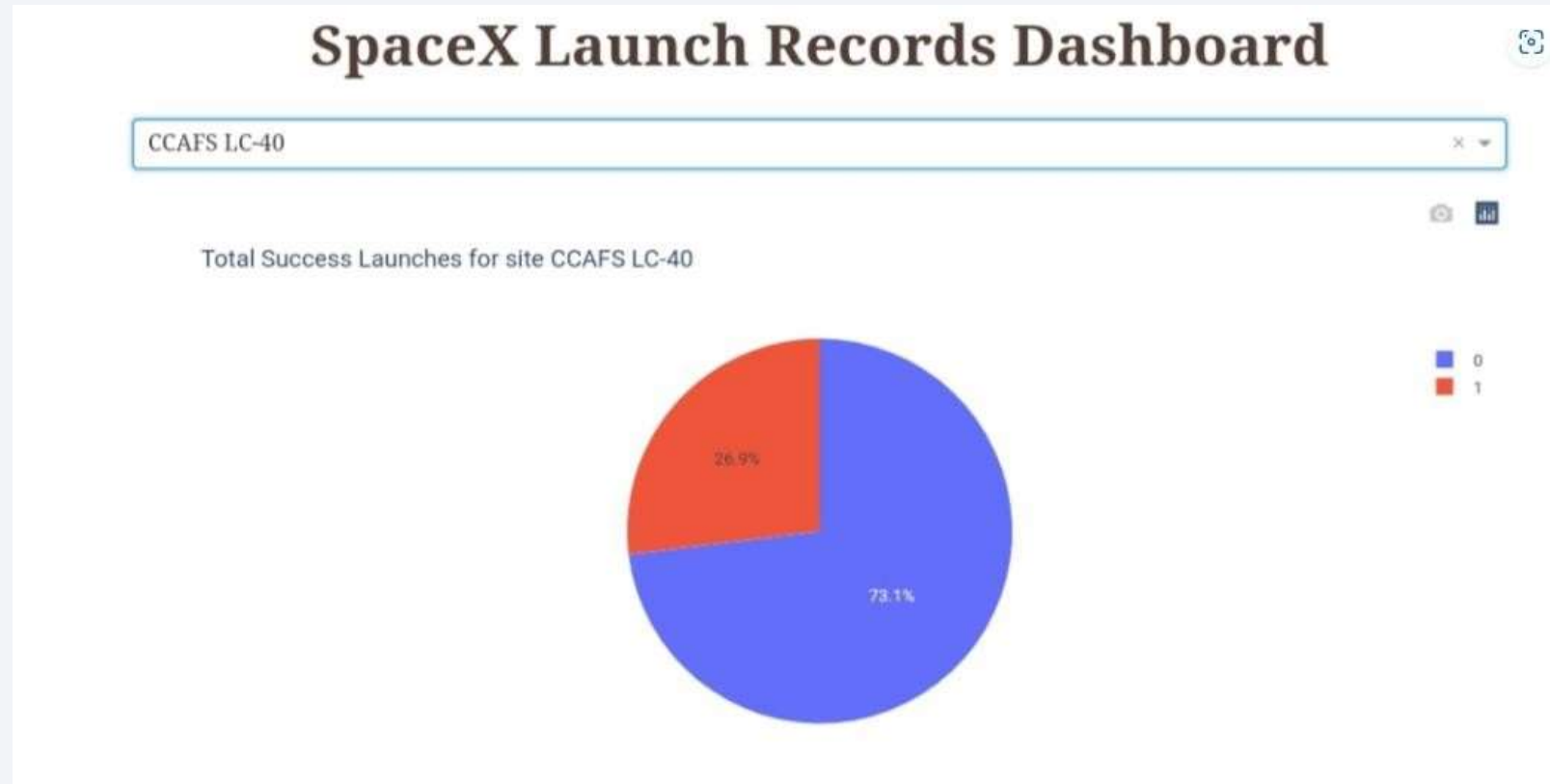
Success Count for all launch sites



- KSC LC 39A: 41.7%
- CCAFS LC-40: 29.2%
- VAFB SLC -4E: 16.7%
- CCAFS SLC-40: 12.5%

The place from where launches are done seems to be a very important factor of success of missions.

Highest success launch ratio



76.9% of launches are successful in this site.

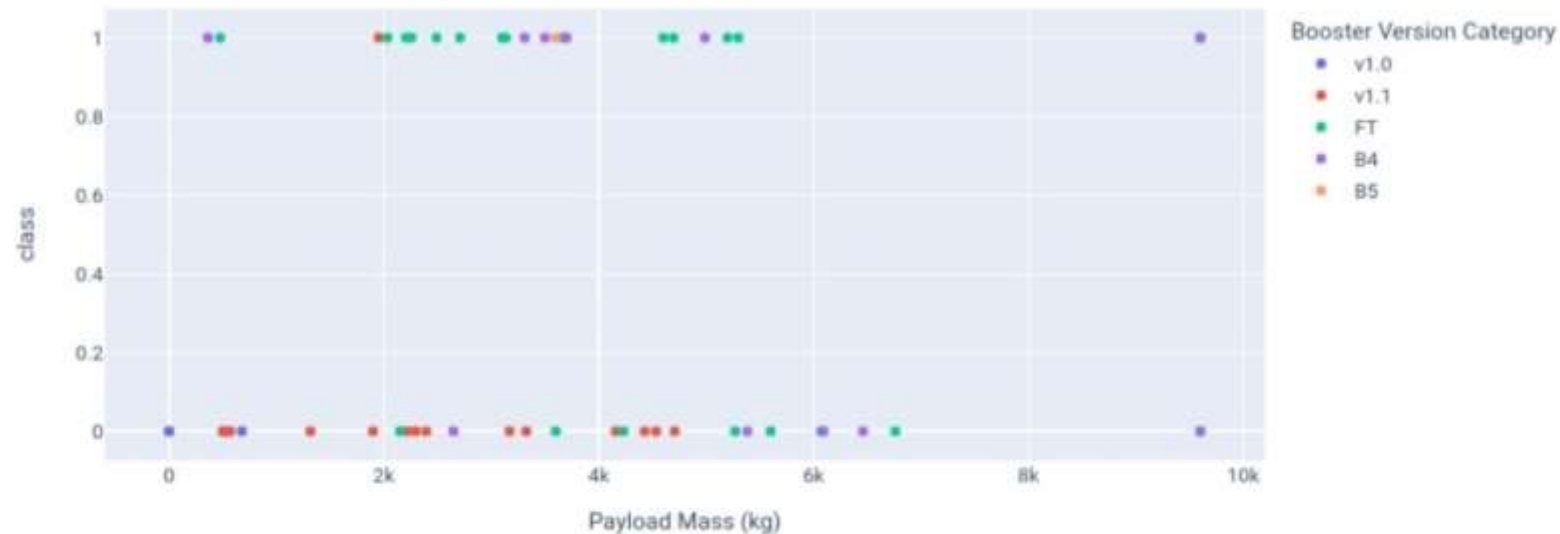
Payload vs. Launch Outcome



Payload range (Kg):



Success count on Payload mass for all sites



Payloads under 6,000kg and FT boosters are the most successful combination.

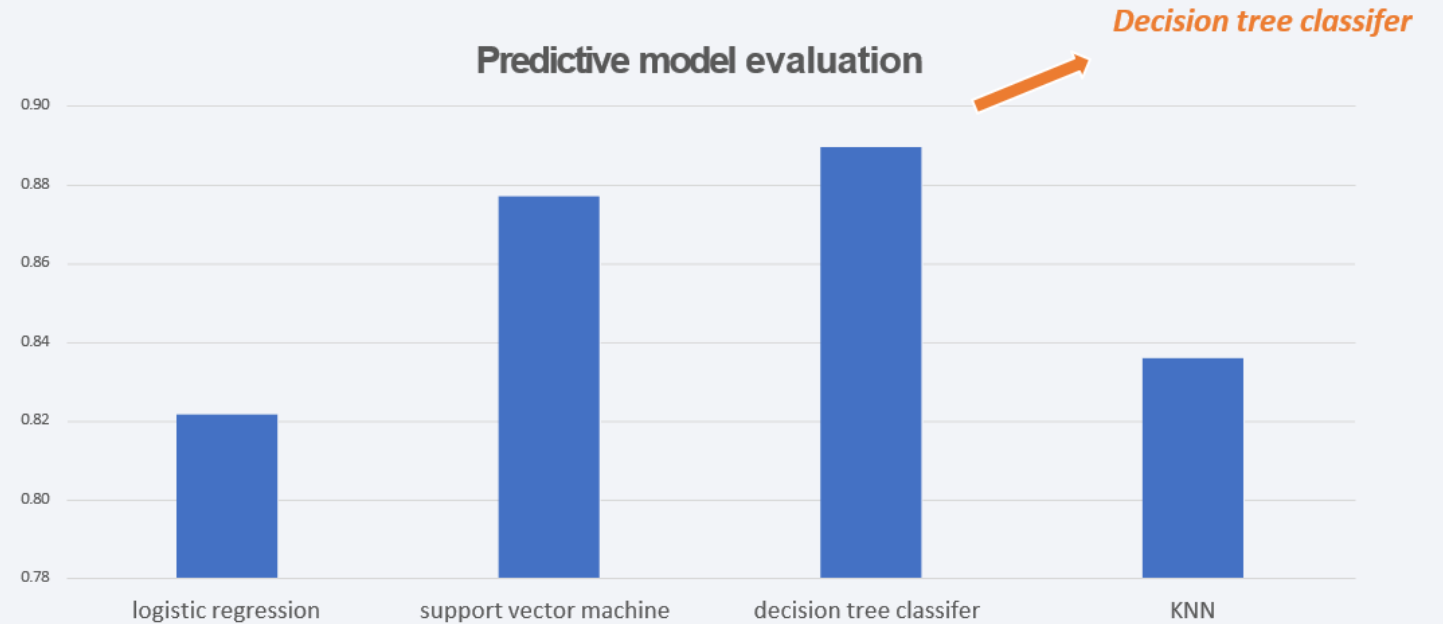


Section 5

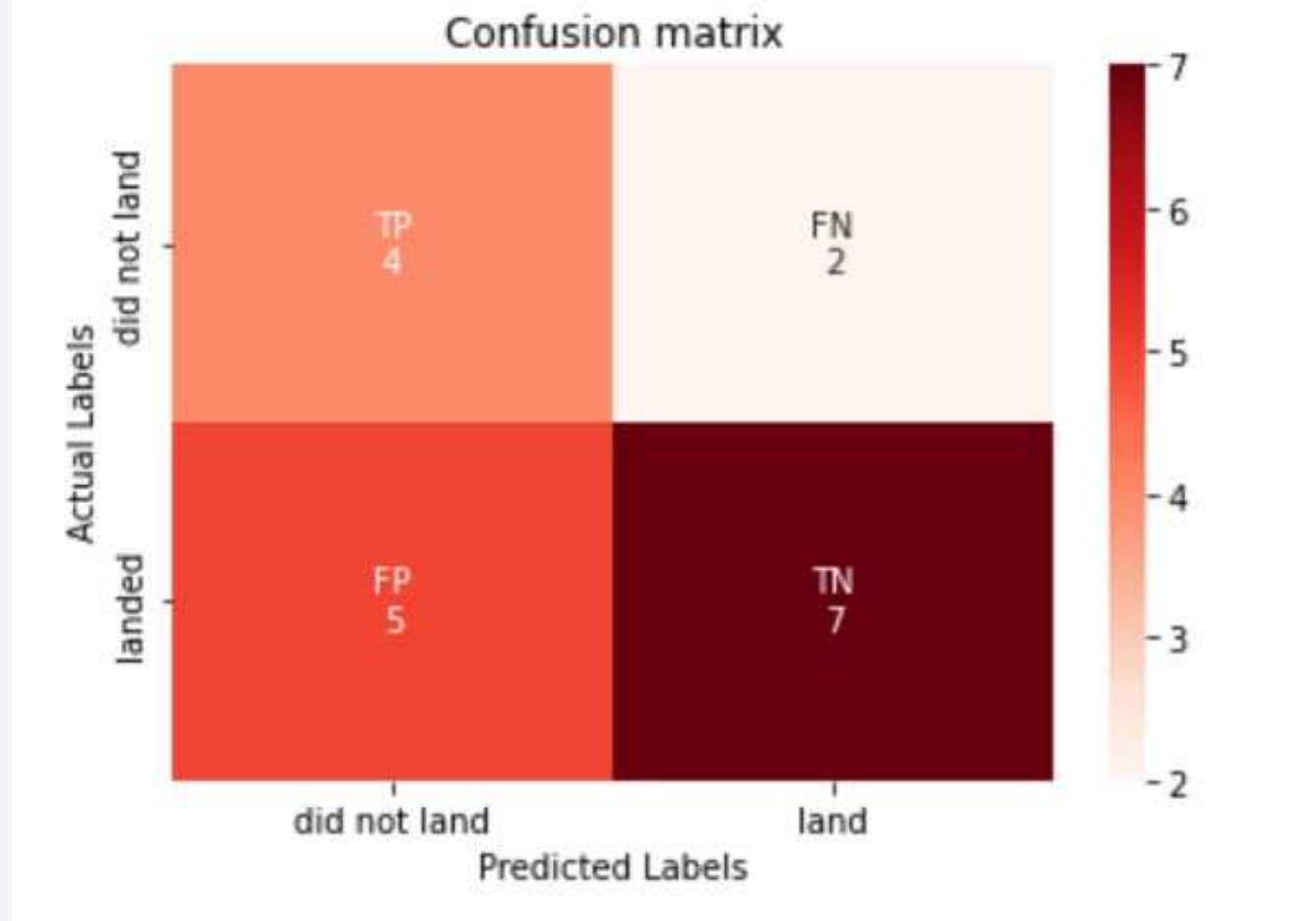
Predictive Analysis (Classification)

Classification Accuracy

- Four classification models were tested, and their accuracies are plotted beside.
- The model with the highest classification accuracy is Decision Tree Classifier, which has accuracies over than 87%.



Confusion Matrix



True Positive = 4

Successful landing correctly predicted.

True Negative = 7

Failure correctly predicted.

False Positive = 5

Failure predicted as successful landing.

False Negative = 2

Successful landing predicted as failure.

Confusion matrix of Decision Tree Classifier proves its accuracy by showing the big numbers Of true positive and true negative compared to the false ones.

Data Insights

- Launch success varied across launch sites and orbit types.
- Payload characteristics were associated with landing outcomes.
- The yearly analysis showed an increasing success trend over the observed period.

Interactive Analytics

- Folium provided geographical analysis of launch sites and nearby infrastructure.
- Plotly Dash enabled interactive exploration of launch site, payload and landing outcomes.

Predictive Analysis

- Four classification algorithms were evaluated.
- The selected model and its performance were evaluated using classification metrics and a confusion matrix.

Overall

The project combined data collection, data wrangling, SQL, visualization, interactive analytics and machine learning to analyze Falcon 9 first-stage landing outcomes.

Creative Enhancements



Interactive Visual Analytics

Created an interactive Folium map to explore launch locations, landing outcomes, and proximity to geographical features such as coast, railway, highway and city.

Interactive Dashboard

Developed a Plotly Dash dashboard with launch-site selection, payload filtering, success-rate visualization and payload/outcome analysis.

Multi-dimensional Analysis

Combined visual EDA, SQL analysis, geographical analysis and machine learning instead of relying on a single analytical approach.

Predictive Modeling

Compared multiple classification approaches — Logistic Regression, SVM, Decision Tree and KNN — and evaluated their performance.

Creative approach: The project combines static analysis with interactive geographical visualization, dashboard-based exploration and predictive modeling to provide multiple perspectives on Falcon 9 landing outcomes.

1. Launch success improved over time

The yearly analysis shows an increasing launch-success trend from 2013 through 2020.

2. Payload and landing outcome are related

The analysis indicates that payload characteristics are associated with first-stage landing outcomes, with successful landings appearing more frequently in certain payload ranges.

3. Orbit type shows different success patterns

Landing success varies by orbit type, with ES-L1, GEO, HEO and SSO showing high observed success rates in the analysis.

4. Geography provides additional context

Launch-site performance can be examined together with geographical proximity to coastlines, railways, highways and cities.

5. Machine learning can predict landing outcomes

Multiple classification algorithms were evaluated, with the Decision Tree reported at approximately 89% accuracy using the selected parameters.

Appendix



- https://github.com/Jananijana2712/SpaceX-Falcon_9_First_Stage_Landing_Prediction
- <https://www.coursera.org/learn/applied-data-science-capstone/home/welcome>

Thank you!

